Task 1 Report: Subdomain and DNS Intelligence Aggregator

Author: Alisha Rokka

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Assessment: 5th Division OSINT, Reconnaissance and Intelligence Automation (Internship Task)

Reference: Task 1

Authorized target: zonetransfer.me (a public lab domain made for DNS testing)

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1. Executive Summary

I built a tool that collects public information about a domain from several sources and saves it in one clean JSON file. It looks up DNS records, WHOIS details, and subdomains listed in Certificate Transparency (CT) logs, then links the subdomains to their IP addresses.

I ran it against zonetransfer.me, a domain designed to be tested. The tool collected **20 DNS records**, found **2 subdomains** through CT sources, and linked those subdomains to the IP address they share. All 13 required features work. Every result carries its source, timestamp, and confidence level, and duplicates are removed.

CNAME and AAAA records did not appear for zonetransfer.me, so I tested them on two other public domains. Both worked.

Main conclusion: the tool meets every Task 1 requirement, stays inside the authorized scope, and keeps verified data separate from inferred data.

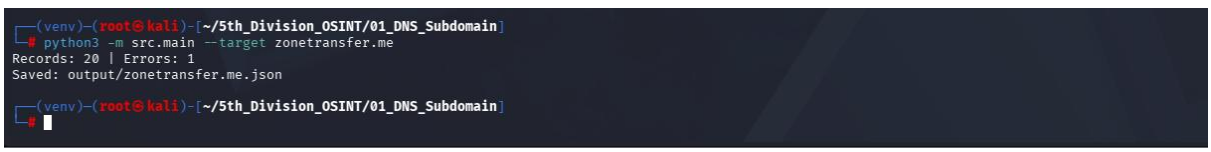
2. Scope and Rules Followed

- **Target:** only zonetransfer.me. Other domains were used only for read-only DNS lookups to demonstrate CNAME and AAAA.
- **Sources:** public DNS, public WHOIS, and public CT logs (crt.sh and Certspotter).
- **No credentials:** no API keys or passwords are used or stored in the code.
- **Safe behavior:** input is validated, results outside the target domain are discarded, requests have timeouts, and failed requests retry with a delay.

3. How the Tool Works

Module	What it does
validators.py	Cleans the domain input. Rejects URLs, emails, and lookalike domains. Checks scope.
dns_enum.py	Collects A, AAAA, MX, NS, TXT, CNAME, and SOA records using dnspython.
sources.py	Finds subdomains from CT logs (crt.sh, with Certspotter as backup).
correlate.py	Resolves subdomains to IPs, raises confidence for names that resolve, and builds IP records.
whois_info.py	Collects WHOIS fields where available.
main.py	Runs everything, removes duplicates, and writes the JSON output.

Run command: python3 -m src.main --target zonetransfer.me



Every record uses the same structure: type, value, source, timestamp, confidence, scope, metadata.

4. Requirements Coverage

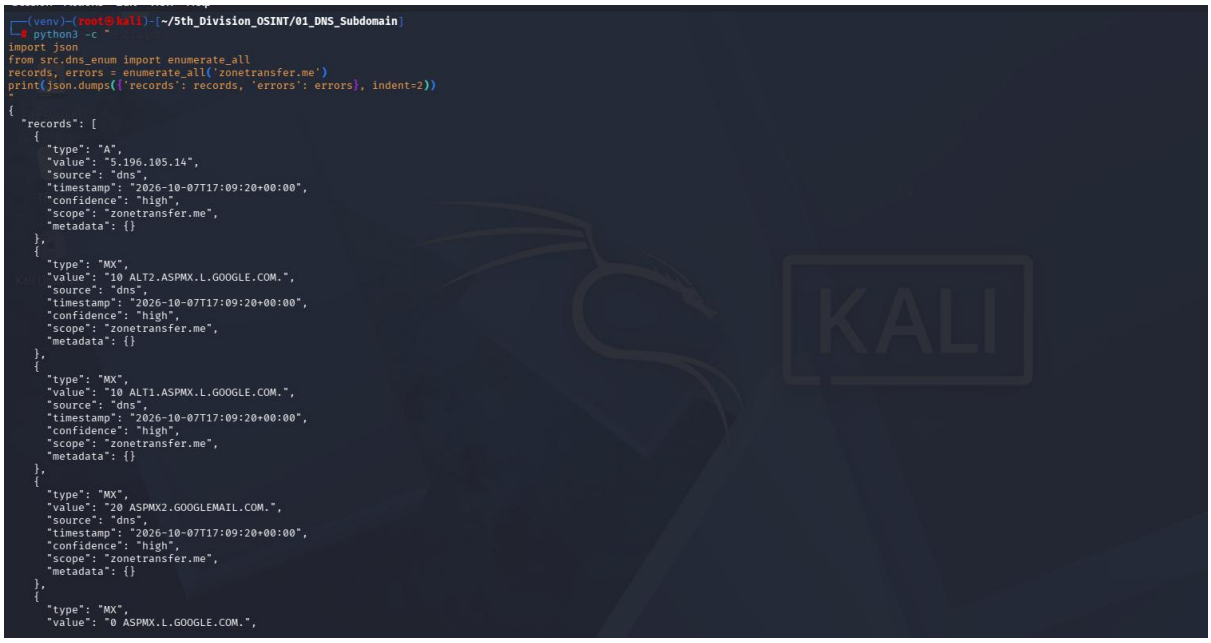
#	**Requirement**	**Status**
1	DNS record enumeration	Done
2	A and AAAA resolution	Done
3	MX identification	Done
4	NS identification	Done
5	TXT collection	Done
6	CNAME identification	Done
7	WHOIS collection	Done
8	CT enumeration	Done
9	Subdomain discovery	Done
10	IP correlation	Done
11	Duplicate removal	Done
12	Source attribution	Done
13	JSON output	Done

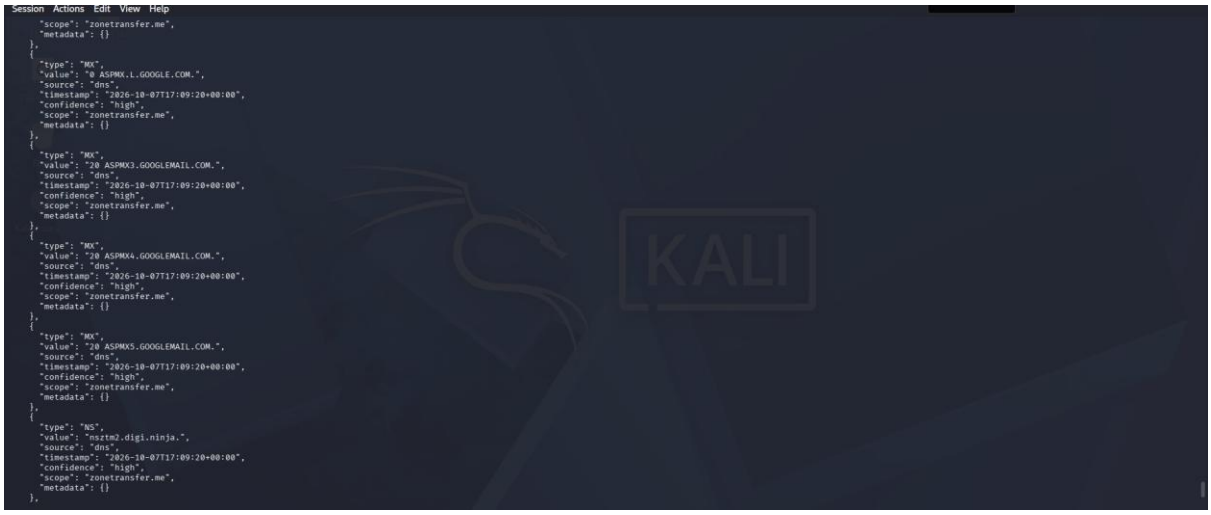
5. Findings

5.1 DNS Records (verified facts)

The tool collected 20 records for zonetransfer.me across A, MX, NS, TXT, and SOA types, with no errors.

- **A record:** 5.196.105.14
- **MX:** mail is handled by Google mail servers
- **NS, TXT, SOA:** collected and stored in the JSON





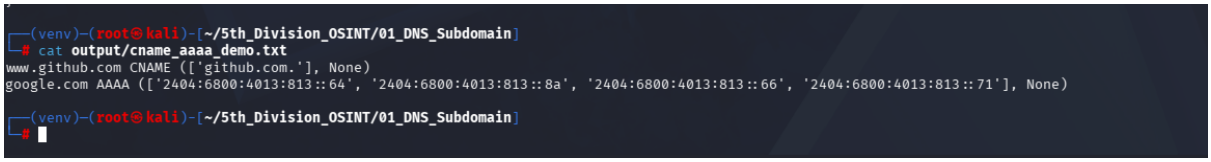


5.2 CNAME and AAAA Verification

zonetransfer.me publishes neither record type, which is normal. To show the code works, I tested two public domains:

- www.github.com CNAME returned github.com.
- google.com AAAA returned several IPv6 addresses (Google rotates these, so they change between runs)

Evidence: output/cname_aaaa_demo.txt



5.3 WHOIS Information

The tool collected the registrar, registration dates, and domain status. WHOIS fields vary by registry, so missing fields are expected and do not cause errors.

Registrar: Mesh Digital Limited

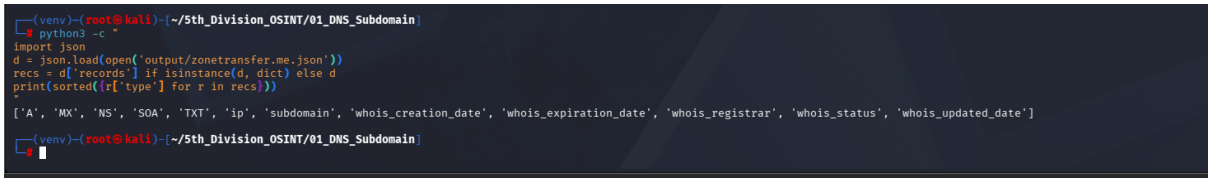
Creation date: 2011-12-27

Expiration date: 2028-12-27

Last updated: 2026-08-11

Status: clientDeleteProhibited, clientRenewProhibited (plus any further status lines in the JSON; only the first two were visible in your output)





5.4 Certificate Transparency and Subdomains

The tool looks for subdomain names in two public certificate lists: crt.sh and Certspotter.

In my test run, crt.sh was down (it returned an HTTP 502 error). The tool did not crash. It saved the error and moved on to Certspotter, which worked. This shows the backup system works.

Certspotter found 2 names:

Name	**What it is**	**Source**	**Confidence from certificate only**	**Confidence after IP check**
www.zonetransfer.me	Subdomain	Certspotter	Medium	High
zonetransfer.me	Main domain	Certspotter	Medium	High

Only www.zonetransfer.me is a real subdomain. The main domain shows up because the same certificate covers both.

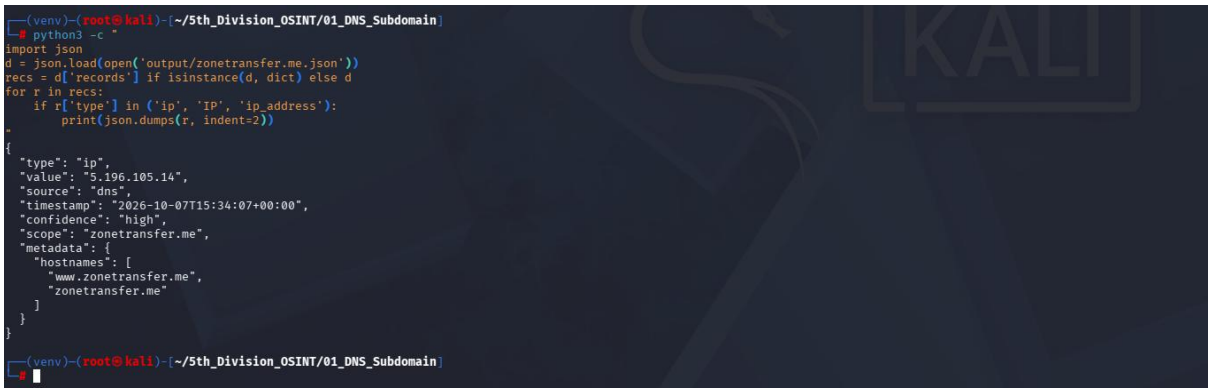
Why the confidence changed: a name on a certificate does not prove the website is live today, so it starts at medium. The tool then checked both names and found they point to the same IP address, 5.196.105.14. Because they are confirmed live, confidence was raised to high.

The tool also removes wildcard prefixes (*.), merges duplicates, and drops any names outside the target domain.



5.5 IP Correlation

Each subdomain was resolved to an IP address. Names that resolved were upgraded in confidence. One IP record links to both hostnames, showing they share the same infrastructure.



5.6 Duplicate Removal and Source Attribution

When a name appears in more than one source, it is merged into one record and all sources are listed in metadata.sources. Every record shows where it came from.

6. Verified Facts vs Inferred Information

Type	Examples
Verified fact	DNS records returned directly by DNS servers; WHOIS fields from the registry
Correlated	A subdomain linked to an IP because it resolved to it
Inferred (lower confidence)	A subdomain found only on a certificate, which is not yet confirmed live

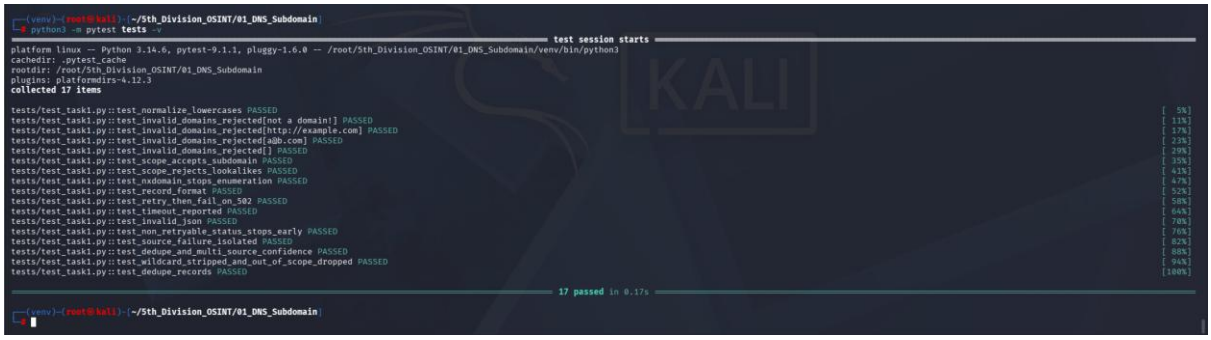
7. Testing

Tests run offline with the network mocked, so they are repeatable. They cover:

- Valid domain resolution
- Invalid and malformed domains (URLs, emails, lookalikes)
- API and network failures
- Timeouts and retries
- Duplicate handling
- Scope checking

Run: python3 -m pytest tests -v

Results saved in: output/test_results.txt



8. Security Controls

- No hard-coded keys or passwords (none are required)
- Domain input validation and strict scope enforcement
- Request timeouts, retries with backoff, and a User-Agent header
- Each module fails safely: one broken source never stops the whole run
- Dependencies pinned in requirements.txt and checked with pip-audit

9. Limitations and Known Issues

- **crt.sh can be slow or unavailable.** The tool retries and falls back to Certspotter, but results can still be incomplete.
- **Certificate names are not proof of live hosts.** That is why those results stay at medium confidence until they resolve.
- **Passive discovery only.** The tool does not brute-force subdomains, so unlisted subdomains will be missed.
- **No Censys or SecurityTrails.** These were optional and need API keys, which were not used.
- **WHOIS output varies** between registries, so some fields may be empty.
- **Results change over time.** For example, Google's AAAA addresses rotate, so reruns can differ.
- **Small test target.** zonetransfer.me has few subdomains, so larger domains were not tested.

10. False-Positive Considerations

- CT names may belong to hosts that no longer exist.
- Shared hosting means two subdomains on the same IP are not necessarily owned by the same party.
- Scope filtering reduces, but cannot fully remove, lookalike-domain noise.

11. Conclusion

The aggregator meets all 13 Task 1 requirements. It collects data from several public sources, removes duplicates, records the source of every result, and produces normalized JSON. It stays within the authorized scope and handles errors safely. The main weaknesses are dependence on third-party CT services and passive-only discovery, both described above.

12. Evidence Index

File	Purpose
output/zonetransfer.me.json	Main JSON output
output/cname_aaaa_demo.txt	CNAME and AAAA proof
output/test_results.txt	Test results
README.md	Setup and usage